

Positive Multilinear Forms on Non-Atomic L_p

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Question 3 from:

Luis A. Garcia, Jose Lucas P. Luiz, Vinicius C. C. Miranda, *Norm attainment for multilinear operators and polynomials on Banach Spaces and Banach lattices*, arXiv:2605.12117.

The source location is `arxiv.tex:1007--1071`, final section “Open Questions”, Question 3. The question appears on page 28 of the source paper:

We showed in Section 3 that all positive bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ are norm-attaining while there are non-norm attaining bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$. From this and our results, we obtain that all positive bilinear operators $\ell_4 \times \ell_4 \rightarrow L_1([0, 1])$ are weakly sequentially continuous while there are bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ that are not weakly sequentially continuous. It would be interesting, if we get examples with the domain being non-atomic Banach lattices. In view of this, we leave the following question:

Question 3: Are all positive n -linear forms on $L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1]$ with $\sum_{i=1}^n \frac{1}{p_i} < 1$ weakly sequentially continuous?

The paper defines an n -linear operator $A : X_1 \times \cdots \times X_n \rightarrow Y$ to be weakly sequentially continuous when

$$A(x_{1,k}, \dots, x_{n,k}) \rightarrow A(x_1, \dots, x_n)$$

whenever $x_{j,k} \xrightarrow{w} x_j$ in X_j for every $j = 1, \dots, n$.

Remark 1 (Exponent convention). *The rendered question writes $L_{p_i}[0, 1]$ and assumes $\sum_i 1/p_i < 1$. We read this in the surrounding finite-exponent convention of the paper, namely $1 < p_i < \infty$. The inequality already excludes $p_i = 1$. A human reviewer should confirm whether the authors intended to include the endpoint $p_i = \infty$; the proof below resolves the finite-exponent formulation.*

Claim

Theorem 1. *Let $n \geq 2$ and let $1 < p_1, \dots, p_n < \infty$ satisfy*

$$\sum_{i=1}^n \frac{1}{p_i} < 1.$$

Then there is a positive continuous n -linear form on

$$L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1]$$

which is not weakly sequentially continuous. In particular, Question 3 has a negative answer under the finite-exponent interpretation.

Proof. Define

$$A(f_1, \dots, f_n) = \int_0^1 f_1(t) \cdots f_n(t) dt.$$

Since $\sum_i 1/p_i < 1$, choose $s \in (1, \infty]$ such that

$$\frac{1}{s} = 1 - \sum_{i=1}^n \frac{1}{p_i}.$$

Holder's inequality, with the remaining factor filled by the constant function $1 \in L_s[0, 1]$, gives

$$|A(f_1, \dots, f_n)| \leq \prod_{i=1}^n \|f_i\|_{p_i}.$$

Thus A is a continuous n -linear form. It is positive in the real Banach-lattice sense, because $f_i \geq 0$ for all i implies $f_1 \cdots f_n \geq 0$ almost everywhere.

Let (r_k) be the Rademacher functions on $[0, 1]$. For every $1 < p < \infty$, one has $r_k \xrightarrow{w} 0$ in $L_p[0, 1]$. Indeed, if q is the conjugate exponent and $h \in L_q[0, 1]$, approximate h in L_q by a dyadic step function h_m . For all sufficiently large k , $\int_0^1 h_m r_k = 0$, and Holder's inequality controls $\int_0^1 (h - h_m) r_k$.

Now set

$$x_{1,k} = r_k, \quad x_{2,k} = r_k, \quad x_{j,k} = 1 \quad (3 \leq j \leq n).$$

Then

$$x_{1,k} \xrightarrow{w} 0, \quad x_{2,k} \xrightarrow{w} 0, \quad x_{j,k} \xrightarrow{w} 1 \quad (3 \leq j \leq n),$$

but

$$A(r_k, r_k, 1, \dots, 1) = \int_0^1 r_k(t)^2 dt = 1$$

for every k , while

$$A(0, 0, 1, \dots, 1) = 0.$$

Therefore A is not weakly sequentially continuous. □

Verification Notes

No computational verification is needed. The proof reduces to three standard facts: Holder's inequality, positivity of the integral form on real L_p lattices, and weak nullity of the Rademacher sequence in finite $L_p[0, 1]$.

The only convention-sensitive point is whether Question 3 intended the endpoint $p_i = \infty$. The packet should be read as a full negative answer for the finite-exponent version used in the surrounding paper.

Human Review: Positive multilinear forms on non-atomic L_p

Original automatic proof: `solution_packet.pdf`

Checked date: 13 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed solution appears to give a correct negative answer to Question 3 from the source paper, under the natural interpretation that the spaces are in the finite-exponent range $1 < p_i < \infty$. The counterexample is the positive n -linear form

$$A(f_1, \dots, f_n) = \int_0^1 f_1(t) \cdots f_n(t) dt$$

on

$$L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1], \quad \sum_{i=1}^n \frac{1}{p_i} < 1.$$

This form is bounded and positive, but it is not weakly sequentially continuous.

Boundedness and Positivity

The boundedness is the standard Holder argument. Since

$$\sum_{i=1}^n \frac{1}{p_i} < 1,$$

one can choose $r \in (1, \infty]$ such that

$$\sum_{i=1}^n \frac{1}{p_i} + \frac{1}{r} = 1.$$

Applying Holder's inequality to $f_1 \cdots f_n \cdot 1$ gives

$$|A(f_1, \dots, f_n)| \leq \|f_1\|_{p_1} \cdots \|f_n\|_{p_n} \|1\|_r.$$

Thus A is a continuous n -linear form. Positivity is immediate: if each $f_i \geq 0$, then the integrand is non-negative almost everywhere.

Failure of Weak Sequential Continuity

Let (r_k) be the Rademacher functions. For every finite p , the sequence (r_k) is weakly null in $L_p[0, 1]$. Hence $r_k \rightarrow 0$ weakly in each relevant finite L_{p_i} -space.

However,

$$A(r_k, r_k, 1, \dots, 1) = \int_0^1 r_k(t)^2 dt = 1$$

for every k , whereas

$$A(0, 0, 1, \dots, 1) = 0.$$

Therefore A is not weakly sequentially continuous.

Interpretation

The argument is short, but it seems to be a genuine obstruction to the stated question. The main point to check is not the calculation itself, but whether the authors intended the question exactly in this finite-exponent form, or whether some additional hypothesis was meant to exclude this Rademacher-product example.

Review Outcome

Archive this packet as an apparently correct full negative answer, pending author feedback. I recommend contacting the authors to ask whether the argument is known to them, whether it matches the intended formulation of the question, and how they would assess the value of the example.